Qassim University جامعة القصيم
College Of Computer | كلية الحاسب

# E-Trek Application


Submitted by:

Shuaa Alrashidi

group with

Hanady Alharby

Aisha Al-harbi

Anfal Al-harbi

Raghad Khulaif Al-Harbi

Aliyah Al-rasheedi

Hessa Alharby

Bayan alotaibi

# table of content :

Content.....Pages
Introduction.....3
Stakeholders.....4
Functional Requirements.....5-6
Non-Functional Requirements.....7-8
Structured Specification of Major Requirements.....9
Scenario.....10
Selecting the Most Suitable Architectural Pattern.....14
Software Architectural Design.....15
Use Case Diagram.....16
the Sequence Diagram.....17
activity Diagram.....18
Class Diagram.....19

# Introduction:

**E-Trek**
**is an intelligent system designed to monitor and analyze vegetation health and combat desertification in the Qassim region using artificial intelligence, remote sensing, satellite imagery, and drones.**
**The system offers an interactive, user-friendly interface that supports data-driven environmental decision-making. It also generates environmental reports powered by language generation models (e.g., LLaMA).**

**Project Contributions:**
- **Empowering environmental agencies to make strategic, data-driven decisions.**
- **Automating satellite image analysis to extract indicators such as NDVI.**
- **Raising public awareness through smart, visual, and easy-to-understand reports.**
- **Focusing initially on the Qassim region, with plans for national expansion.**

# Stakeholders

- **Environmental authorities and government entities: are responsible for monitoring desertification and tracking the progress of afforestation initiatives across different regions.**
- **Local farmers: benefit from accessing up-to-date information about vegetation status and recommended strategies for improving land conditions.**
- **Environmental researchers: use the data to analyze changes in vegetation and understand patterns of soil desertification over time.**
- **The general public and local communities: are engaged through smart maps and environmental reports, helping raise awareness and encourage participation in environmental preservation.**
- **AI models: are employed to analyze satellite imagery and generate insightful recommendations for afforestation efforts.**
- **Project supervisors: oversee the implementation of afforestation projects and track their progress in each city to ensure goals are being met.**

# Functional Requirements (1-2)

| RQ.ID | Requirements | Category |
|---|---|---|
| FR-01 | Allow users to recover deleted data within a specific time. | Data Recovery |
| FR-02 | Provide internal statistical analysis tools. | Data Analysis |
| FR-03 | Generate interactive charts for data visualization. | Data Visualization |
| FR-04 | Integrate live weather data with satellite imagery. | Data Integration |
| FR-05 | Allow users to customize the dashboard. | User Interface |
| FR-06 | Provide an API for external integration. | System Integration |
| FR-07 | Alert users when data errors are detected. | System Monitoring |
| FR-08 | Allow exporting reports in multiple formats (CSV, JSON). | Reporting |

# Functional Requirements (2-2)

| RQ.ID | Requirements | Category |
|---|---|---|
| FR-09 | Automatically update environmental data from external sources. | Automation |
| FR-10 | Upload satellite or drone images for analysis. | Core Functionality |
| FR-11 | Perform AI-based analysis on uploaded imagery. | Core Functionality |
| FR-12 | Select the type of analysis (NDVI or object detection). | Core Functionality |
| FR-13 | Visualize analysis results in interactive maps. | Data Visualization |
| FR-14 | Filter cities by vegetation index (NDVI) in descending order. | Data Filtering |
| FR-15 | Compare desertification between two time periods. | Comparative Analysis |
| FR-16 | Generate AI-based Arabic reports summarizing findings. | Reporting |

# Non-Functional Requirements (1-2)

| RQ.ID | Requirements | Category |
|---|---|---|
| NFR-01 | The system homepage must load within 2 seconds. | Performance |
| NFR-02 | Each satellite image must be processed and analyzed within 5 minutes. | Performance |
| NFR-03 | The system must handle at least 50 analysis requests per hour without performance degradation. | Performance |
| NFR-04 | Environmental reports must be generated within 30 seconds of user request. | Performance |
| NFR-05 | The system must support secure login with optional Two-Factor Authentication (2FA). | Security |
| NFR-06 | Unauthorized users must not be able to access private reports or data of other organizations. | Security |
| NFR-07 | All unauthorized login attempts must be logged and reported to the system administrator. | Security |
| NFR-08 | A new user must be able to understand and navigate the basic features within 10 minutes without training. | Usability |

# Non-Functional Requirements (2-2)

| RQ.ID | Requirements | Category |
|---|---|---|
| NFR-09 | The user interface must support at least two languages ( Arabic and English). | Usability |
| NFR-10 | The system must support accessibility tools for people with disabilities, such as text zoom and high-contrast mode. | Usability |
| NFR-11 | The system must function properly on Chrome,  Edge, and Safari browsers. | Compatibility |
| NFR-12 | The interface must be responsive, supporting screens from mobile phones to wide monitors. | Compatibility |
| NFR-13 | The system must be scalable to support up to 10,000 active users without requiring major architectural changes. | Scalability |
| NFR-14 | The system must have a minimum uptime of 99.5% during official working hours. | Reliability |
| NFR-15 | In case of a failure, the system must automatically restart or notify the support team within 5 minutes. | Reliability |
| NFR-16 | The system’s architecture must be easy to modify and maintain by the development team. | Maintainability |

# Structured Specification of Major Requirements

| Requirement | Function | Description | Inputs | Outputs | Action |
|---|---|---|---|---|---|
| FR1 | Upload satellite data for processing | System receives compressed satellite image data for analysis. | .zip file (e.g., B4, B5, B7 bands) | File is uploaded and ready for processing. | User uploads a .zip file of satellite bands. |
| FR2 | Select analysis method | System determines the kind of environmental analysis to apply. | Analysis type (NDVI or YOLO) | Selected analysis method initialized. | User selects type of analysis (NDVI or Object Detection) |
| FR3 | Analyze image using AI model | Image is analyzed using the selected AI model. | Uploaded image bands | NDVI score or object detection result | System runs selected model (NDVI or YOLO). |
| FR4 | Display analysis result | Final results are visualized and displayed to the user. | AI model output | Heatmap / vegetation index / object map | System returns results as visualizatio n. |

# Scenario

**Scenario 1:**

- **Monitoring Plant Health Using Satellites and Drones**

| | |
|---|---|
| **Main** | The user uploads satellite images and drone-captured images in Qassim. The system analyzes the images using AI techniques to extract indices such as NDVI and SAVI to assess plant health. The results are displayed in the form of maps showing the level of vegetation health. |
| **Description** | • The user uploads satellite or drone-captured images in supported formats such as JPEG, PNG, or GeoTIFF.<br>• The system compresses the images to improve performance and speed up the analysis.<br>• The system analyzes the images using artificial intelligence to extract plant indices like NDVI and SAVI.<br>• The results are displayed in the form of interactive maps that show the health level of the plant.<br>• The user views the results through a customizable dashboard.<br>• The user can export the analysis results in multiple formats, such as CSV or JSON. |
| **Alternative** | 1. If the uploaded images are of poor quality or unclear:<br>a. The system asks the user to upload higher-quality images.<br>b. A tutorial is provided to guide the user on how to capture proper satellite or drone images.<br>2. If the uploaded file format is not supported:<br>a. The system shows an error message and lists the supported formats.<br>3. If an error occurs during analysis:<br>a. The system alerts the user and logs the issue internally for future tracking.<br>4. If live weather data cannot be integrated:<br>a. The system continues with the analysis without it and informs the user. |

**Scenario 2:**

- **Generating an Intelligent Report in Arabic**

| | |
|---|---|
| **Main** | **After performing the analysis, the user selects "Arabic" to generate the report. The results are passed to the pre-trained LLaMA model to generate a report that includes environmental analysis, recommendations, challenges, and areas for improvement in the studied area.** |
| **Description** | • After reviewing the plant health analysis, the user selects Arabic as the language for the report.<br>• The system sends the analysis results to a trained language model to generate a report.<br>• The report includes key sections such as:<br>  ◦ Environmental analysis<br>  ◦ Recommendations<br>  ◦ Identified challenges<br>  ◦ Suggestions for improvement in the area |
| **Alternative** | 1. If the system fails to generate the report in Arabic:<br>  a. The report is generated in English as a fallback.<br>  b. The user is notified that the Arabic version could not be created.<br>2. If there is a technical issue with the language model or API:<br>  a. The system tries again or offers a simpler version of the report.<br>3. If the user wants to save the report:<br>  a. It can still be downloaded in formats like CSV or JSON, regardless of the language. |

**Scenario 3:**

- **Filter cities by highest vegetation coverage**

| | |
|---|---|
| **Main** | **When the user selects the "Display cities by vegetation coverage" feature, the system will sort and rank cities within Qassim according to their NDVI index, arranging them from highest to lowest. The results can be presented as either a detailed table or a visual heat map.** |
| **Description** | • User Action:<br>User selects the "Display cities by vegetation coverage" feature from the menu/interface.<br>• System Action:<br>The system retrieves the latest NDVI data for all cities in the Qassim region.<br>• System Action:<br>The system filters out any cities with missing NDVI data (to handle in the alternative flow).<br>• System Action:<br>The cities with available NDVI data are sorted in descending order (highest NDVI to lowest).<br>• System Action:<br>The system generates output.<br>A heat map showing NDVI by city.<br>• User Action:<br>The user selects output ( map).<br>• System Action:<br>The system displays the selected output ( heat map).<br>• Postcondition:<br>The user can interpret NDVI-based vegetation coverage across Qassim cities and optionally export or share the results. |
| **Alternative** | 1.Missing NDVI Data for Some Cities:<br>• if NDVI data is missing for some cities.<br>The system excludes those cities from the ranking.<br>The system displays a note in the map legend.<br>“Data collection is underway” for cities without NDVI data.<br>2.Missing NDVI Data for All Cities:<br>• if no cities have NDVI data.<br>The system displays a message to the user.<br>“NDVI data is currently unavailable. Please check back later as data collection is ongoing.”<br>The process is terminated without generating heat map.<br>3.System Error or Timeout:<br>• If there is a failure in retrieving data due to a network or backend issue.<br>The system displays an error.<br>“Unable to retrieve NDVI data at this time. Please try again later.”<br>Logging may be triggered for administrative review. |

## Scenario 4: • Presentation of changes in desertification over the years

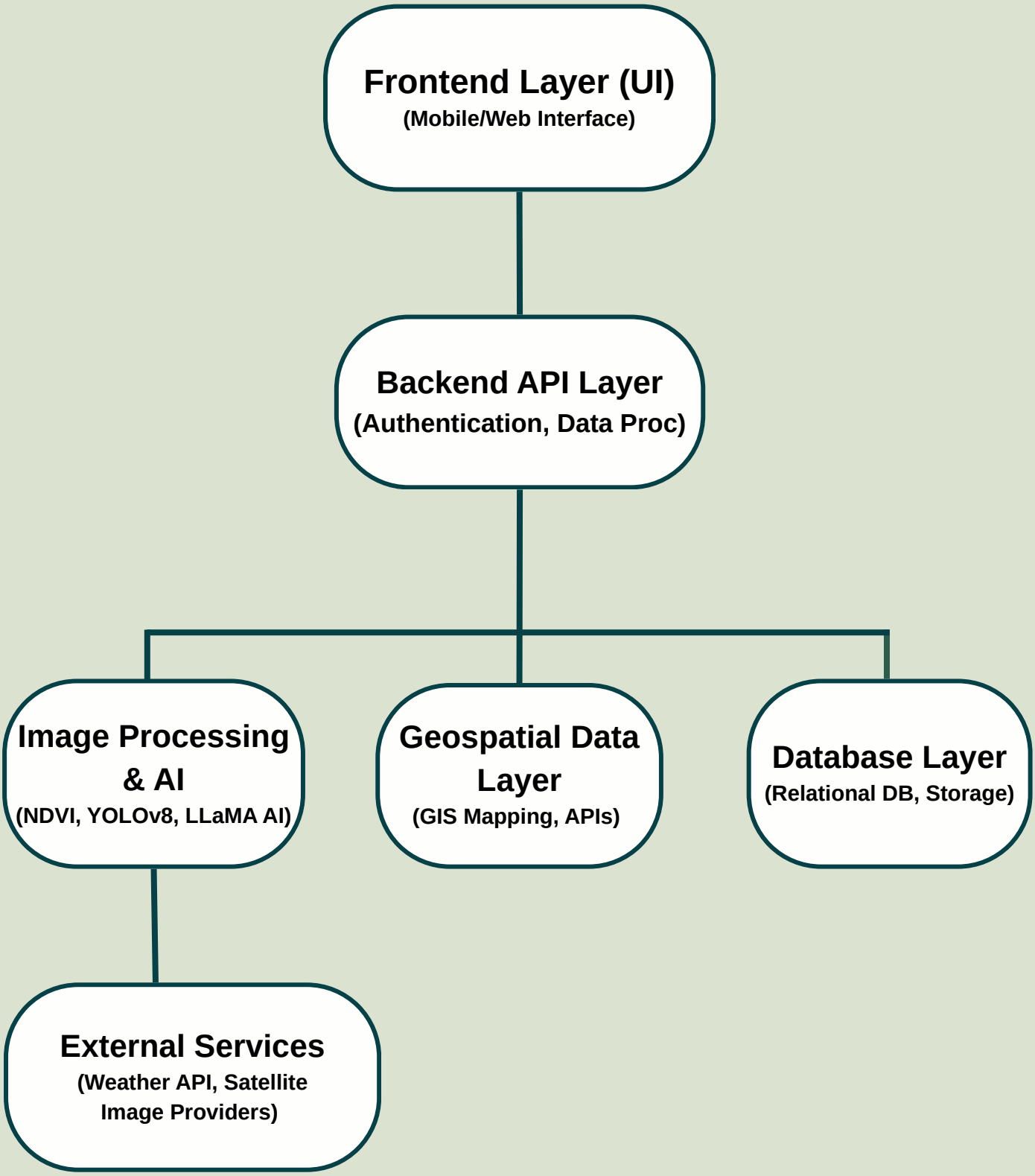
| | |
|---|---|
| **Main** | **The user selects a city like Buraydah and asks to see the state of desertification between two years, such as 2020 and 2025. The system presents a time chart of plant status and desertification compared to previous years, highlighting critical points.** |
| **Description** | • User Action : The user selects a specific city (e.g., Buraydah) and specifies a time range (e.g., 2020–2025).<br>• System Response : The application retrieves historical data on plant health and desertification trends for the selected city and time period.<br>• Visualization : The system generates a time chart that compares the current desertification status with previous years. This chart includes:<br>  ○ Plant status indicators (e.g., vegetation density, green cover percentage).<br>  ○ Desertification metrics (e.g., soil erosion rates, sand encroachment).<br>  ○ Critical points are highlighted, such as significant increases or decreases in desertification levels.<br>• Insights Provided :<br>  ○ Trends in desertification over the specified years.<br>  ○ Comparison with historical data to identify patterns or anomalies.<br>  ○ Identification of areas where interventions might be needed based on the observed changes. |
| **Alternative** | 1. If sufficient data are not available for one year, the system handles this scenario gracefully by providing an alternative visualization and alerting the user about the data gap. Here's how it works:<br>2. Data Gap Detection :<br>  a. The system checks the availability of complete data for each year within the specified time range.<br>  b. If data for a particular year is incomplete or missing, the system identifies this issue.<br>3. Visualization Adjustment :<br>  a. For years with incomplete data, the system displays a sporadic line on the time chart instead of a continuous trend line.<br>  b. The sporadic line is visually distinct (e.g., dashed or dotted) to indicate that the data for that year is not fully reliable.<br>  c. An "incomplete data" note is added near the sporadic line to explicitly inform the user about the data gap.<br>4. User Alert :<br>  a. A user alert message is shown at the top of the chart or in a modal window. The message reads something like:<br>  b. "Note: Data for [specific year] is incomplete. Please consider this when interpreting the results."<br>  c. The alert provides an option for the user to re-search or update data from another source . This option could include:<br>  i. A button labeled "Search for Additional Data" that redirects the user to a search interface where they can input keywords or filters to find supplementary data.<br>  ii. A link to external databases or APIs that might have more comprehensive data for the missing year.<br>  iii. A prompt to upload their own data if they have access to additional sources.<br>5. Fallback Option :<br>  a. If no alternative data source is available, the system continues to display the sporadic line and incomplete data note but ensures that the rest of the chart remains functional and interpretable.<br>  b. The system may also suggest focusing on years with complete data for a more accurate analysis.<br>6. Enhanced User Experience :<br>  a. The system ensures that the user is not left confused by the incomplete data. By providing clear visual cues and actionable options, the user can make informed decisions about how to proceed.<br>  b. The alert message is designed to be non-intrusive but noticeable, ensuring that users are aware of the data limitations without disrupting their workflow. |

# Selecting the Most Suitable Architectural Pattern

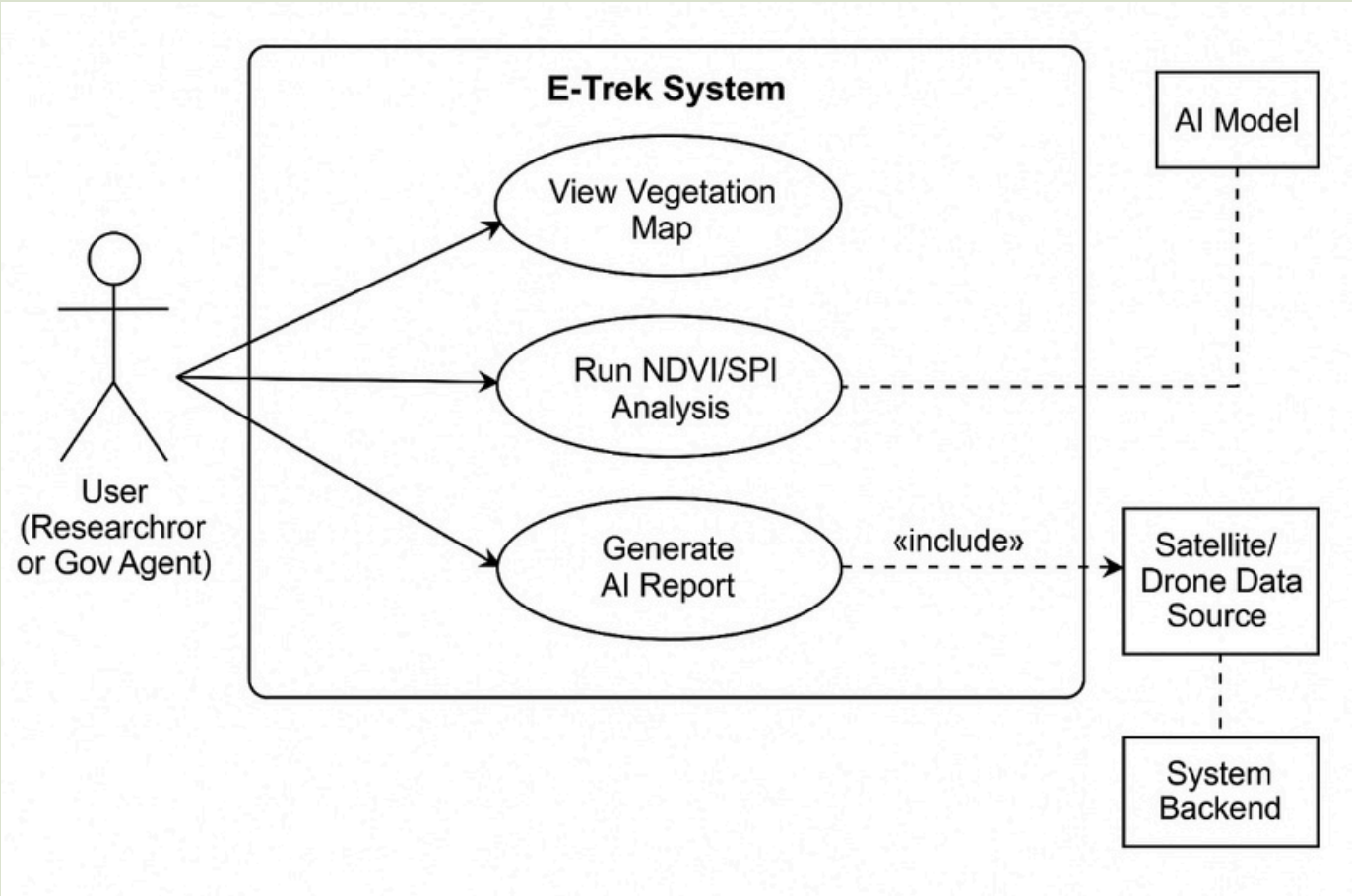
The most suitable architectural pattern for our project E-Trek**" is the Layered Architecture"**
We selected this pattern because it divides the system into well-structured layers that separate the user interface business logic, and data access. This approach improves maintainability, enhances scalability, and simplifies future development.
In the context of E-Trek, the system can be divided into**:**
Presentation Layer**:** Contains the web/mobile interface where users upload images, request reports, and view results.
Business Logic Layer**:** Handles all processing and AI based analysis of satellite and drone images.
Data Layer**:** Manages the storage of environmental data analysis results, and geospatial information.
External Services Layer**:** Includes external APIs such as satellite image providers or weather data sources.

**This** pattern provides a clear organizational structure and allows efficient collaboration between team members during both development and maintenance phases.

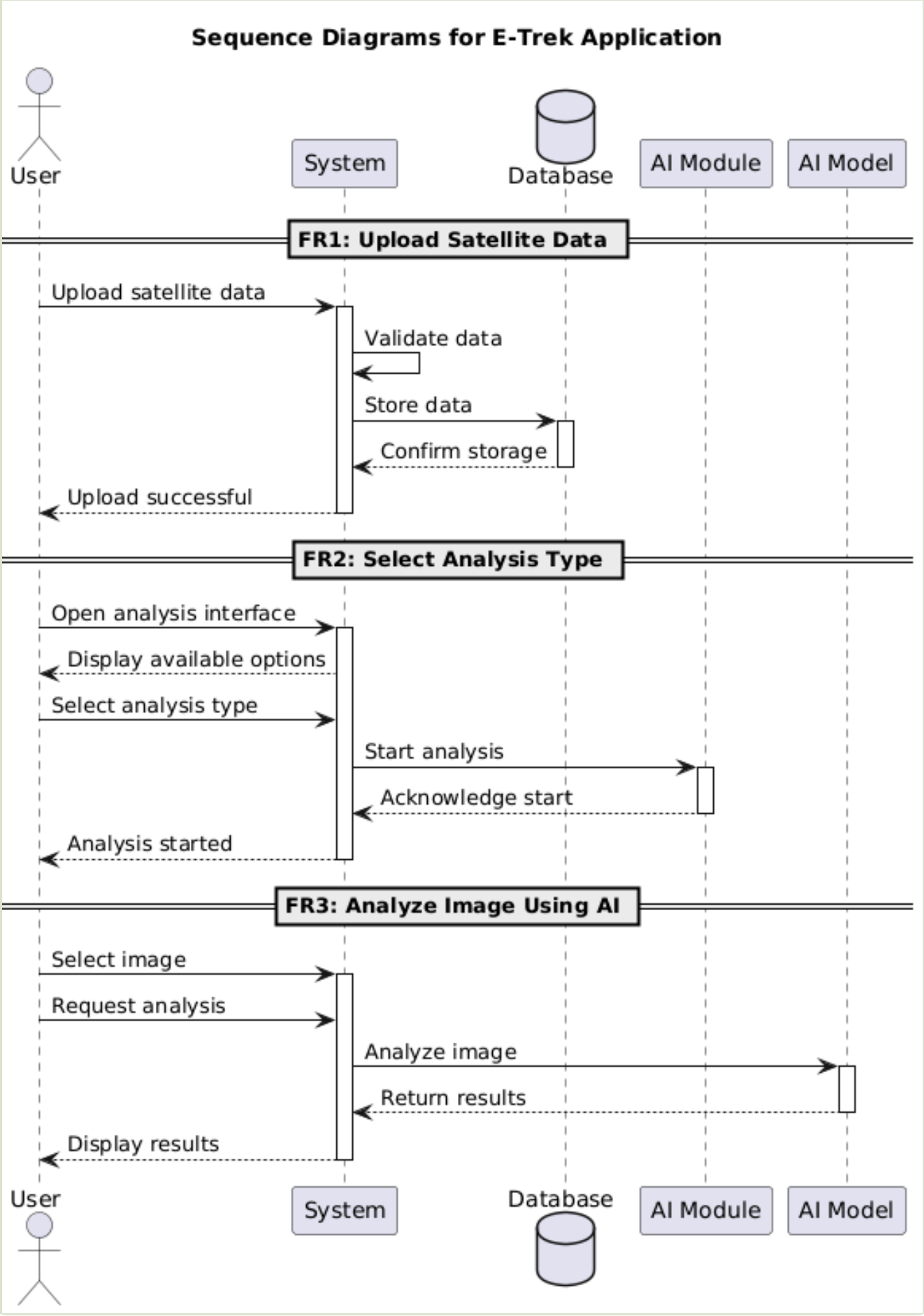
# Software Architectural Design



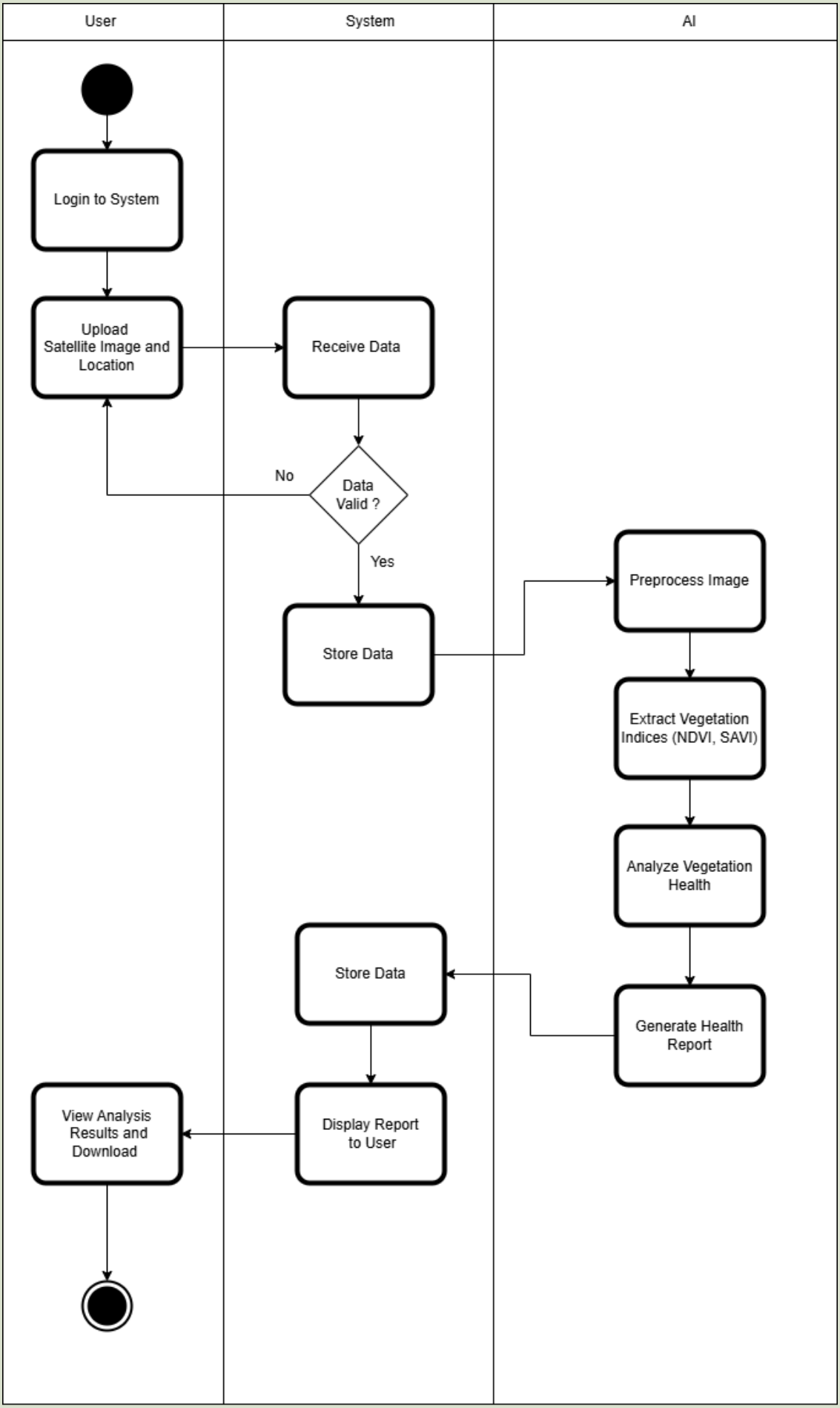
# Use Case Diagram



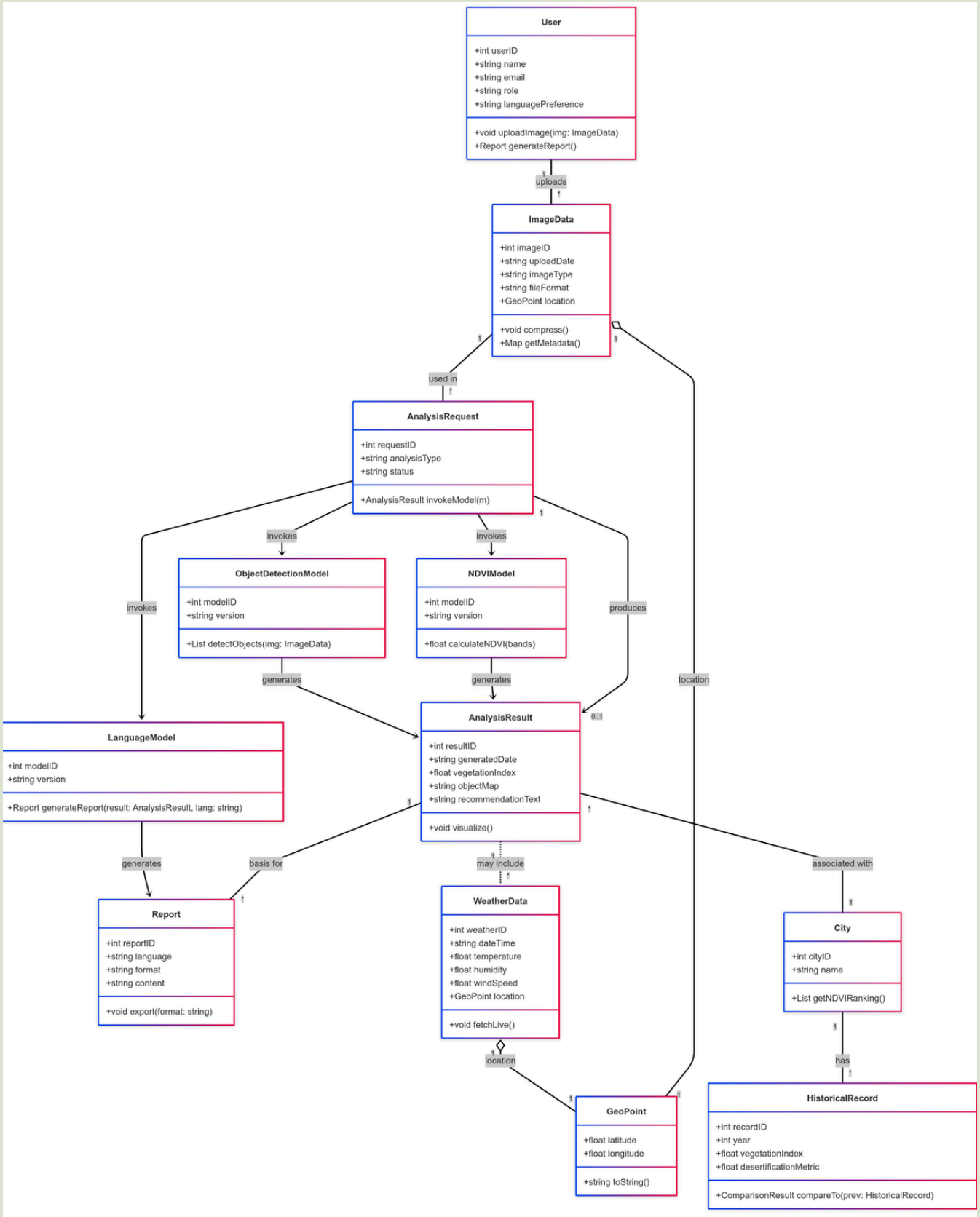
# the Sequence Diagram



# activity Diagram

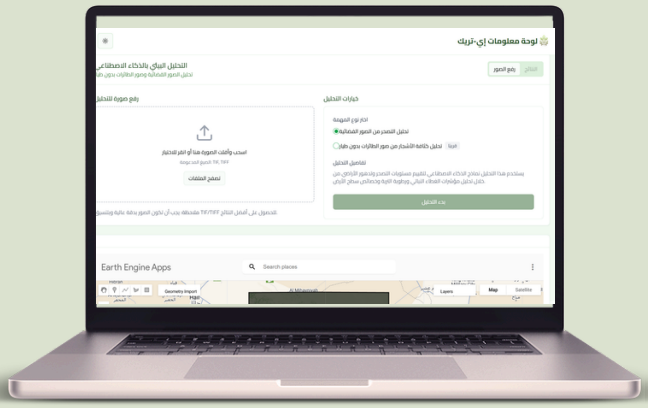


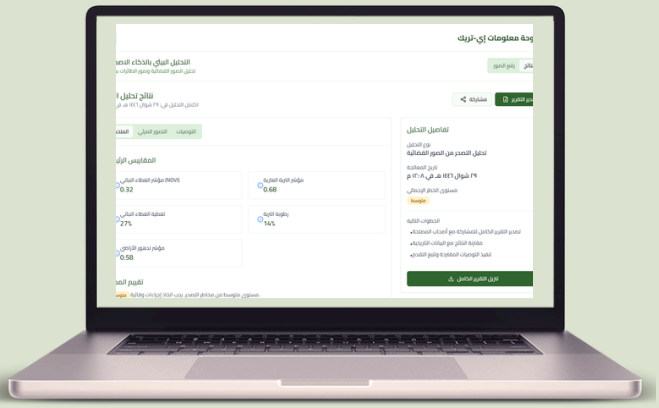
# Class Diagram



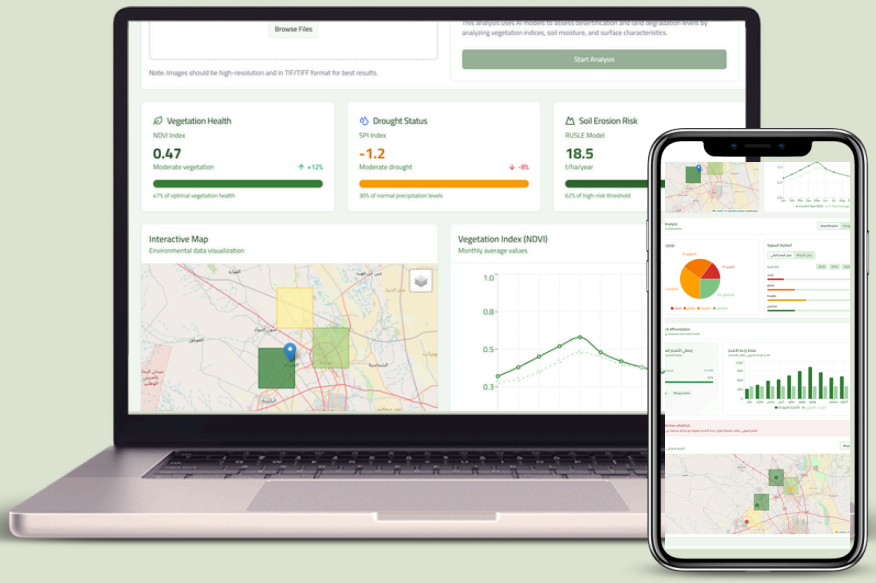
# نموذج اولي

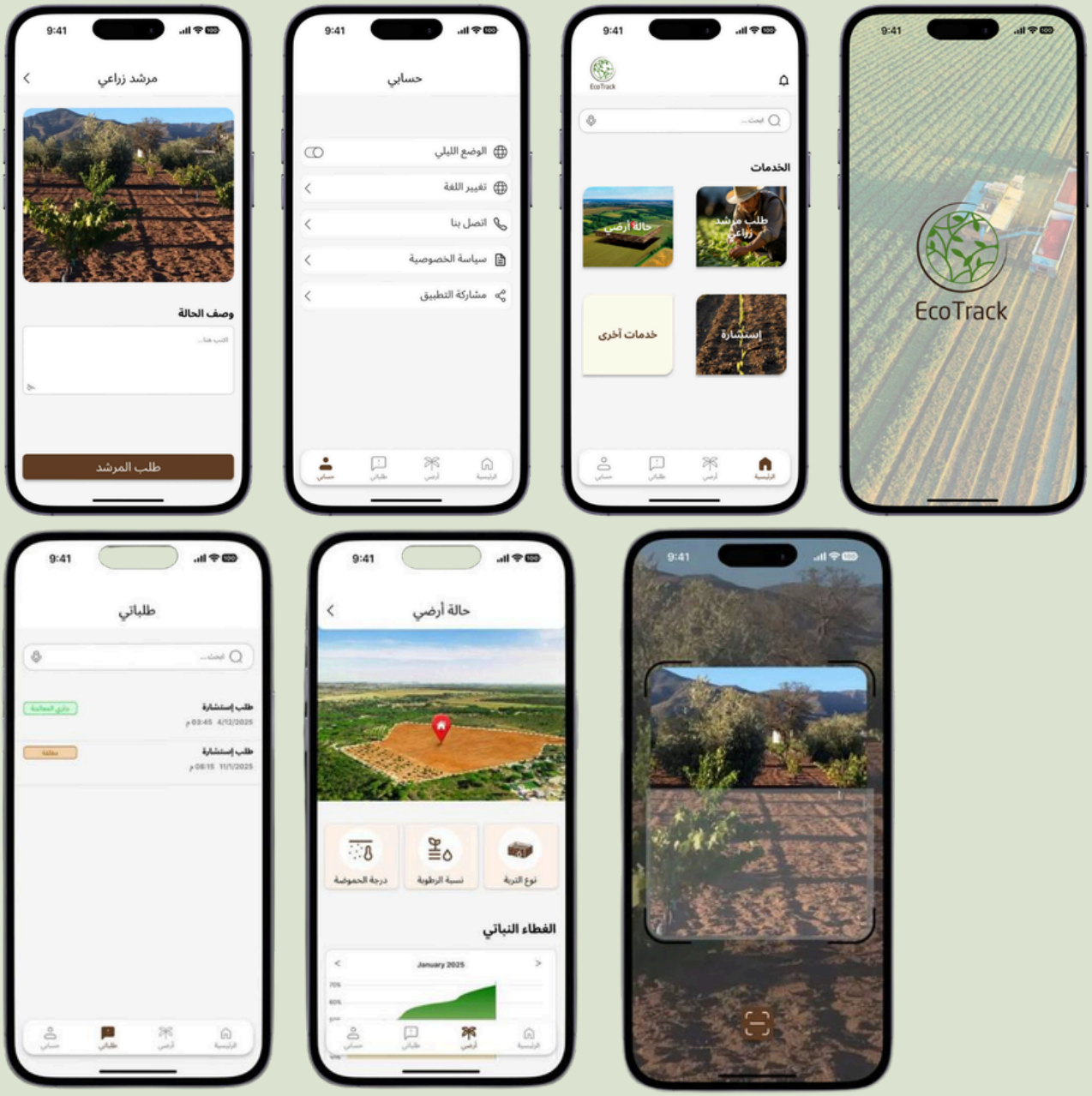
## رفع الصور ونتائج الذكاء الاصطناعي




## Dashboard






# قريبًا:

# إطلاق تطبيق E-Trek الذكي

**شكرًا لحسن استماعكم**
**ونأمل أن يكون خطوة فاعلة**
**نحو بيئة ذكية ومستدامة**